- The equation for single spherical surface is, $\frac{n}{v}-\frac{1}{u}=\frac{(n-1)}{R}$
- The focal length of the thin lens is positive for a converging lens and negative for a diverging lens. It is not based on the position of the focal point.
- The lens makers formula is, $\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
- The lens equation is, $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$
- The magnification produced by the lens is, $m=\frac{h'}{h}=\frac{v}{u}$, $m=\frac{h'}{h}=\frac{f}{f+u}$ (or) $m=\frac{h'}{h}=\frac{f-v}{f}$
- The *power* of a lens is a measure of the degree of convergence (or) divergence of light falling on it. Power and focal length are inverse to each other.
  $P=\frac{1}{f}$
- The unit of power is diopter (D) when the focal length is taken in meter.
- The effective focal length of lenses in contact is, $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$
- A prism produces deviation on the incident ray.
- Angle of deviation depends on angle of prism, angle of incidence and refractive index of material of prism given by the equation, $d=i_1+i_2-A$
- At minimum deviation, $i_1 = i_2$, $r_1 = r_2$, and the ray inside the prism is parallel to the base of the prism.
- The refractive index of prism depends on angle of prism and angle of minimum deviation given by the equation, $n=\frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$
- When white light travels through a medium, different colours travel with different speeds leading to dispersion of light. Red colour travels faster than violet colour in a medium. In vacuum all the colours travel with the same speed.
- The angle of deviation produced by the small angled prism is, $\delta=(n-1)A$
- The angular separation between the two extreme colours (violet and red) in the spectrum is called angular dispersion.
  $\delta_V - \delta_R = (n_V - n_R)A$
- Dispersive power is the measure of ability of the medium to disperse white light.
  $\omega=\frac{(n_V-n_R)}{(n-1)}$
- Rainbow is formed by dispersion of light by droplets of water.
- The scattering of light by particles of size less than wavelength of light is called Rayleigh scattering. The intensity of light produced by Rayleigh scattering is, $I \propto \frac{1}{\lambda^4}$
- Non-Rayleigh scattering is by suspended dust particles whose size is greater than the wavelength of light. This scattering is independent of wavelength of light.

XII_Physics_Vol-2_UNIT-6_EM.indd 46 26-12-2022 11:38:43